

Individual Assignment 1
Due at the beginning of the class on Feb. 04
Total: 5 points

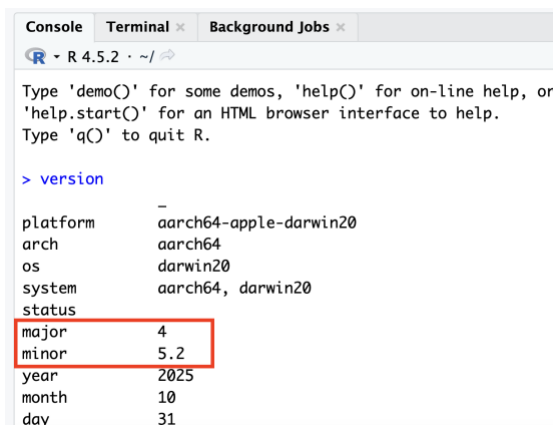
Let us use data analytical skills to determine which physiochemical properties make a wine good!

The dataset is related to white variants of the Portuguese "Vinho Verde" wine. "Vinho verde" is a unique product from the Minho (northwest) region of Portugal. Medium in alcohol, is it particularly appreciated due to its freshness (especially in the summer)!

This assignment asks you to use the dataset "wine.csv" where "quality" is the outcome/target variable. Use R to analyze the data.

Paste your plots generated in RStudio to Word, answer all the questions in Word, and upload it to Canvas. **Please paste your R codes in your Word document as an Appendix for partial credits.**

1. **Verify your R version.** Open RStudio and type "version" in the Console, as shown in the figure below. Confirm that you are using R-4.5.2. Take a screenshot of your Console output and paste the screenshot into your submission file.



```
Console Terminal Background Jobs
R 4.5.2 ~ /

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> version

platform      aarch64-apple-darwin20
arch          aarch64
os            darwin20
system        aarch64, darwin20
status
major          4
minor          5.2
year          2025
month          10
day            31
```

2. Use ggplot to generate a scatterplot to explore the relationship between **alcohol content** and **wine density**. Put alcohol on the x-axis.
3. Partition the data into **75%** Training and **25%** Validation with a seed of 555, i.e. `set.seed(555)`. Use the training set to run a multiple linear regression for “quality” as the target, vs. all other columns as predictors. Paste the result of the fitted linear regression, i.e. the output of `summary(wine.lm)`.
4. How do you interpret the effect of “volatile acidity” based on the estimates?
5. What is the RMSE for the training set? You need to report 3 decimal digits. (You can use `options(scipen=999, digits = 3)` to format outputs.)
6. Apply the model to the **validation** set and assess the performance. What is the RMSE for the validation set?